

The transshipment problem

Formulation

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
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Formulation

The transshipment problem, also known as the min-cost flow problem, is probably the most typical network optimization problem.

In this video, we provide its formulation.

Motivation



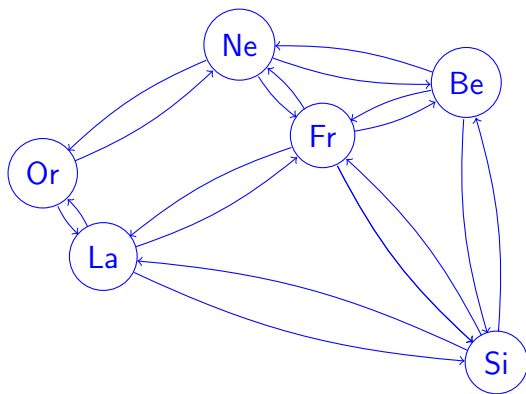
Logistics

- ▶ Goods have to be transported
 - ▶ from the warehouses,
 - ▶ to the clients.
- ▶ They can be transhipped anywhere on the network.
- ▶ Transportation costs must be minimum.

Data

Nodes

- Supply (warehouses).
- Demand (customers).
- Transshipment.



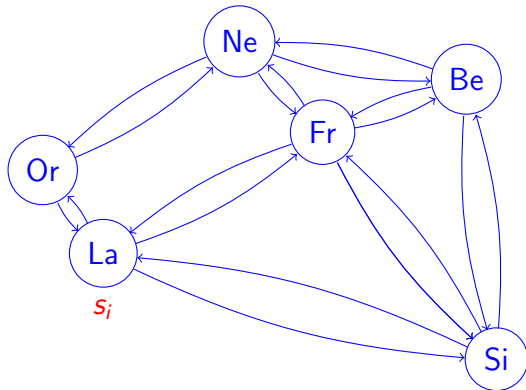
Data

Supply/demand

- ▶ $s_i > 0$ if supply node.
- ▶ $s_i < 0$ if demand node.
- ▶ $s_i = 0$ is transshipment node.

Validity

$$\sum_{i \in \mathcal{N}} s_i = 0.$$



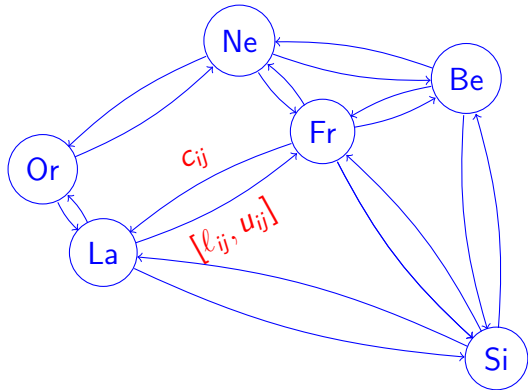
Data

Links

- Costs.
- Capacities.

Validity

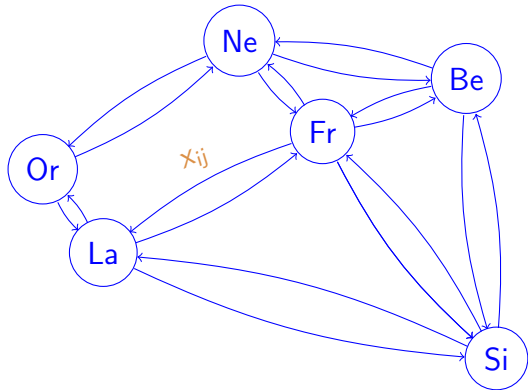
$$\ell_{ij} \leq u_{ij}.$$



Decision variables

Flow vector

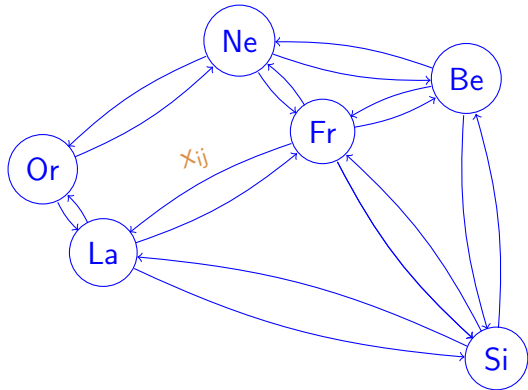
$$x \in \mathbb{R}^n.$$



Objective function

Total costs

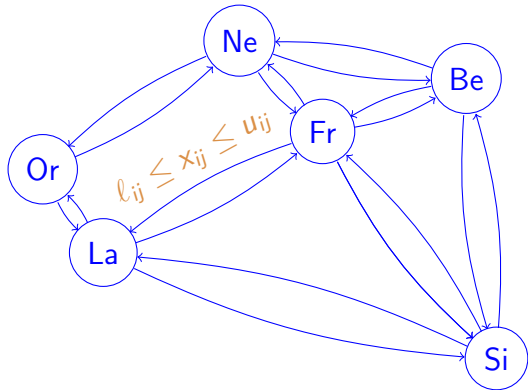
$$\sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}.$$



Constraints

Capacity

$$\ell_{ij} \leq x_{ij} \leq u_{ij}, \forall (i,j) \in \mathcal{A}.$$

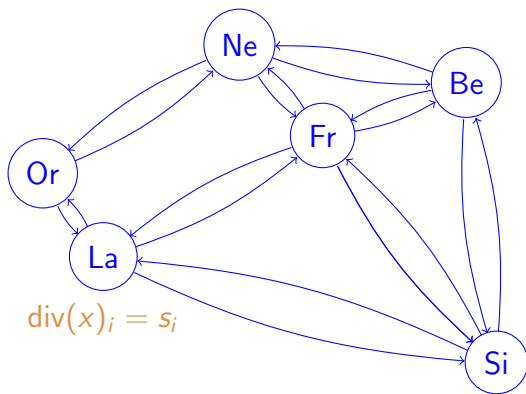


Constraints

Flow conservation

$$\text{div}(x)_i = s_i, \forall i \in \mathcal{N}.$$

$$\sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = s_i, \forall i \in \mathcal{N}.$$



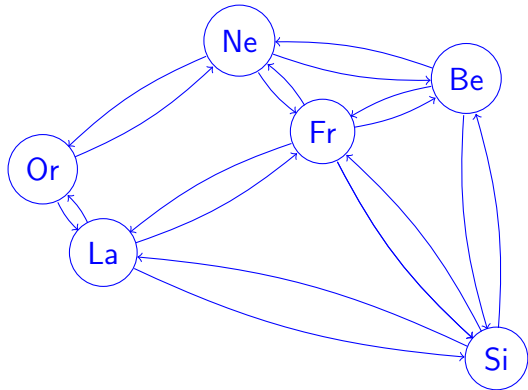
Linear optimization problem

$$\min_{x \in \mathbb{R}^n} \sum_{(i,j) \in \mathcal{A}} c_{ij} x_{ij}$$

subject to

$$\sum_{j|(i,j) \in \mathcal{A}} x_{ij} - \sum_{k|(k,i) \in \mathcal{A}} x_{ki} = s_i, \forall i \in \mathcal{N}.$$

$$\ell_{ij} \leq x_{ij} \leq u_{ij}, \forall (i,j) \in \mathcal{A}.$$



Summary

The transshipment problem consists in finding a flow vector of minimal cost, consistent with supply and demand requests, as well as the network capacity. It is formulated as a linear optimization problem.

This problem allows to solve way more problems than the transshipment problem used to motivate it.

As we will see, it has interesting properties.

Moreover, several important network optimization problems can be expressed as special cases of the transshipment problem.

Consequently, they inherit its interesting properties.